DIGITAL TWIN FOR SEMICONDUCTOR HEALTH MONITORING AND LIFETIME PROTECTION

Project Proposal / Initial Design Document

Project domain	VLSI, Semiconductor Devices, Digital Systems, and Predictive Monitoring
Document purpose	Describe the problem, proposed approach, system architecture, and a practical prototype plan.
Project status	Proposed design. Implementation and experimental results should be added after the prototype is built and tested.

Overview

A semiconductor device can gradually degrade because of temperature, electrical stress, switching activity, and operating conditions. This project proposes a digital-twin approach: a software-based model of a chip or test circuit that receives operating data, estimates its condition, and displays health indicators and warnings. The initial prototype can use simulated data before being connected to real hardware.

Important: This document describes a proposed student project. It does not claim that a working semiconductor digital twin or validated lifetime predictor has already been completed.

1. Problem Statement

Semiconductor circuits may experience performance changes or failure when exposed to excessive temperature, voltage/current stress, or long operating periods. If changes are not detected early, a system may become unreliable or require unexpected maintenance. The project explores how a digital representation of a device, combined with monitoring data and simple health rules, could help identify abnormal operating conditions and support preventive action.

2. Aim and Objectives

• Create a software model representing the operating condition of a semiconductor device or test circuit.

• Generate or collect sample operating data such as temperature, supply voltage, current, and switching activity.

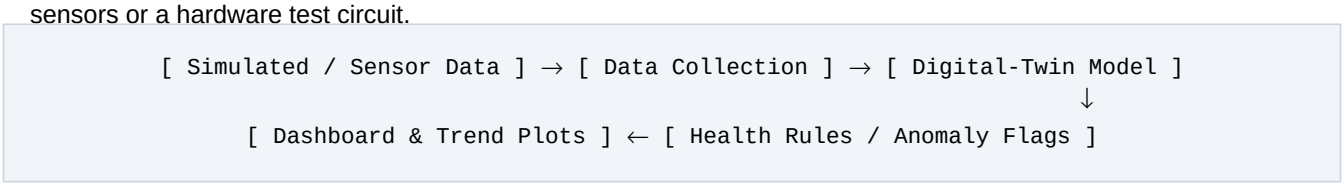
• Compare measured or simulated values with defined safe operating limits.

• Display health status, trends, and warning messages through a simple dashboard or plots.

• Explore a basic degradation or remaining-useful-life estimate only if suitable data and a justified model are available.

3. Proposed System

The first version will use simulated or manually entered data. A monitoring module supplies values to the digital-twin model. The model checks the values against configured limits and calculates a simple status such as Normal, Warning, or Critical. A dashboard presents the latest readings and history. Later versions may connect sensors or a hardware test circuit.



4. Working Principle

Step 1 — Input: Provide sample voltage, current, temperature, or switching-activity values.

Step 2 — Model: Update the software representation with the latest operating values.

Step 3 — Evaluate: Compare values with project-defined thresholds and look for unusual changes.

Step 4 — Report: Show readings, trends, and status messages on the dashboard.

Step 5 — Improve: Validate the rules with test cases and, where possible, measured data.

5. Tools and Requirements

Category	Suggested tools / items
Programming	Python for simulation, data handling, and plots
Model / circuit	A simplified circuit model or synthetic sample dataset; VHDL may be added for a digital test block
Visualisation	Matplotlib or a lightweight dashboard such as Streamlit (optional)
Hardware (optional)	Temperature sensor or test board only if safely available; initial version can be software-only
Documentation	GitHub repository, README, source code, sample data, and test results

6. Implementation Plan

Phase 1 — Define: Choose the device or test circuit and decide which parameters to monitor.

Phase 2 — Simulate: Create a small sample dataset with normal and abnormal operating cases.

Phase 3 — Build the model: Write code that updates the model and checks operating limits.

Phase 4 — Visualise: Plot parameter trends and display a clear health status.

Phase 5 — Test: Run normal, over-limit, missing-data, and boundary-value test cases.

Phase 6 — Document: Upload source code, instructions, sample data, screenshots, and honest test results to GitHub.

7. Expected Outcome

The intended outcome is a working proof-of-concept that represents selected semiconductor operating parameters, flags values that cross chosen limits, and displays simple trends. The prototype can demonstrate the monitoring concept, but reliable lifetime prediction would require appropriate device data, a validated degradation model, and experimental evaluation.

8. Limitations and Safety

• Threshold-based warnings are not the same as a validated fault diagnosis or lifetime prediction.

• Synthetic data can demonstrate software behaviour but cannot prove real-world semiconductor reliability.

• Any hardware testing must remain within the device and board ratings; do not intentionally overheat or over-stress components.

• Report measured results separately from simulated examples.

9. Future Scope

Possible extensions include real sensor integration, a VHDL-based digital test module, anomaly detection, temperature and power trend analysis, comparison across operating conditions, and a more rigorous lifetime-estimation model based on trusted device data.

10. Suggested GitHub Repository Structure

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Digital-Twin-Semiconductor-Health-Monitoring/
■■■ README.md
■■■ docs/
■   ■■■ project-proposal.pdf
■■■ src/
■■■ data/
■■■ tests/
■■■ results/
```

Add code, data, screenshots, and results as they become available. Keep the README clear about which parts are proposed, simulated, or tested on hardware.

Prepared as an initial project document for development and GitHub documentation.